Train a Human Neural Network KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

The rule. total = ($A \times$ Attack weight) + ($S \times$ Special weight) + bias. More than 0: **FIGHTING**. 0 or less: **PSYCHIC**.

The fix rule. Right: no change. Said P, really F: **ADD** A, S, 1. Said F, really P: **SUBTRACT** A, S, 1.

Pass 1. Starting weights: Attack −1, Special +1, bias 0.

Card	A	S	Total	Says	Really	Right?	Attack w	Special w	Bias
T1 Machop	3	1	−2	P	F	no	2	2	1
T2 Abra	1	4	11	F	P	no	1	−2	0
T3 Machamp	4	2	0	P	F	no	5	0	1
T4 Espeon	2	4	11	F	P	no	3	−4	0
T5 Hitmonlee	4	1	8	F	F	yes	3	−4	0
T6 Kirlia	1	2	−5	P	P	yes	3	−4	0
T7 Conkeldurr	5	2	7	F	F	yes	3	−4	0
T8 Alakazam	2	5	−14	P	P	yes	3	−4	0
T9 Makuhita	2	1	2	F	F	yes	3	−4	0
T10 Mr. Mime	2	3	−6	P	P	yes	3	−4	0

Mistakes in pass 1: 4

Pass 2. Same cards, same order. Start with your weights from the end of pass 1.

Card	Total	Says	Right?
T1 Machop	5	F	yes
T2 Abra	−13	P	yes
T3 Machamp	4	F	yes
T4 Espeon	−10	P	yes
T5 Hitmonlee	8	F	yes

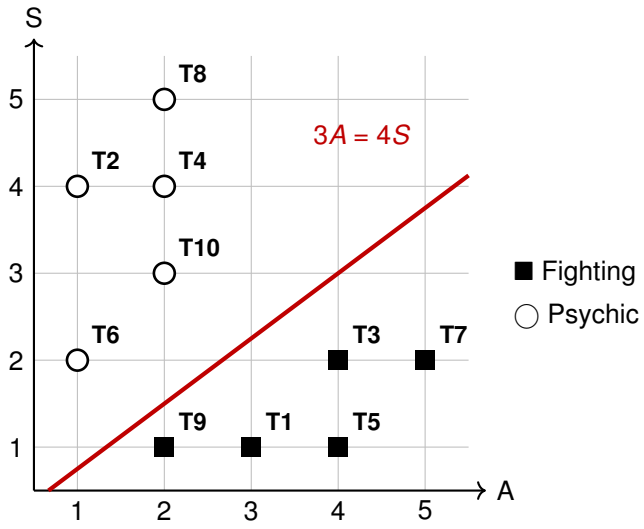
Card	Total	Says	Right?
T6 Kirlia	−5	P	yes
T7 Conkeldurr	7	F	yes
T8 Alakazam	−14	P	yes
T9 Makuhita	2	F	yes
T10 Mr. Mime	−6	P	yes

Mistakes in pass 2: 0 (If you made mistakes, do a pass 3 on the back.)

Test cards. Do **not** fix the weights on a test! Final weights: Attack 3 Special −4 bias 0

Card	A	S	Total	Says	Really (flip the card)
X1 Primeape	4	2	4	F	Fighting
X2 Mewtwo	4	5	−8	P	Psychic
X3 Hypno	2	2	−2	P	Psychic
X4 Gallade	4	2	4	F	Psychic and Fighting
X5 Lucario	4	4	−4	P	Fighting and Steel
X6 Medicham	2	2	−2	P	Fighting and Psychic

The neuron's line. The training cards are on the graph. Draw the line where the total is 0.



Talk about it. The neuron said Gallade is Fighting. Gallade is Psychic **and** Fighting. Was the neuron wrong?

The neuron **was** / **was not** wrong about Gallade, because Answers vary.